Nonlinear Optimization: Homework #2

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1 Convex Functions

1st Problem

25. ***Least-squares problems with uncertainty.*** A large number of applications can be formulated as least-squares problems. A least-squares (LS) problem is an optimization problem of the form

$$\min_{x \in \mathbf{R}^n} \|Ax - b\|^2,$$

where $A \in \mathbf{R}^{m \times n}$ and $b \in \mathbf{R}^m$ are given. So, a least-squares problem seeks the x that minimizes the squared-norm of the residual $Ax - b$.

The canonical form above assumes that both A and b are certain; that is, they are known without ambiguity. In many applications, however, both A and b are uncertain: we may know only that $A \in \mathcal{A}$ and $b \in \mathcal{B}$, where $\mathcal{A}$ is a given bounded set of $m \times n$ matrices, and $\mathcal{B}$ is a given bounded set of m-dimensional vectors. (As an illustrative example, we may know only that each entry of b is in given interval, say, $b_i \in [\alpha_i, \beta_i]$, for $i = 1, 2, \ldots, m$; in this example, we would have $\mathcal{B} = \{b = (b_1, \ldots, b_m) : \alpha_i \leq b_i \leq \beta_i \text{ for all } i\}$.)

In the presence of such uncertainty, we may solve

$$\min_{x \in \mathbf{R}^n} \underbrace{\sup \left\{ \|Ax - b\|^2 : A \in \mathcal{A},\, b \in \mathcal{B} \right\}}_{f(x)}.$$

The rationale here is that, now, we seek the x that minimizes the *worst* squared-norm of the residual, that is, the largest squared-norm of the residual $Ax - b$, as A and b vary over their uncertainty sets $\mathcal{A}$ and $\mathcal{B}$.

Show that this is a convex problem; that is, show that the function f is convex.

Note that both $\mathcal{A}$ and $\mathcal{B}$ have not been specified; you must show that f is convex for any given bounded sets $\mathcal{A}$ and $\mathcal{B}$ (which may or may not be convex).

Solution to the 1st Problem. In order to prove that f is convex, let us start by examining the structure of the objective. Let $\mathcal{I} := \mathcal{A} \times \mathcal{B}$ denote the index set of all admissible pairs (A, b). For each fixed $(A, b) \in \mathcal{I}$, define the function

$$\phi_{A,b} : \mathbf{R}^n \to \mathbf{R}, \quad \phi_{A,b}(x) = \|Ax - b\|^2. \tag{1}$$

We claim that $\phi_{A,b}$ is convex for each fixed $(A, b) \in \mathcal{I}$. To see this, note that $\phi_{A,b}$ can be decomposed as

$$\phi_{A,b} = g \circ L_{A,b}. \tag{2}$$

The two components are defined as follows.

- The function $g : \mathbf{R}^m \to \mathbf{R}$ is the squared Euclidean norm

$$g(y) = \|y\|^2 = y^T y. \tag{3}$$

  Since g is twice differentiable with gradient $\nabla g = 2y$ and Hessian

$$\nabla^2 g = 2I_m \succ 0, \tag{4}$$

  where I_m denotes the $m \times m$ identity matrix, Theorem (Second-Order Criterion) implies that g is convex.

- The map $L_{A,b} : \mathbf{R}^n \to \mathbf{R}^m$ is defined by

$$L_{A,b}(x) = Ax - b. \tag{5}$$

  This is an affine map, since it has the form $L_{A,b}(x) = \ell(x) + w$, where $\ell : \mathbf{R}^n \to \mathbf{R}^m$ is the linear map $\ell(x) = Ax$ and $w = -b \in \mathbf{R}^m$.

Since g is convex, $L_{A,b}$ is affine, and $L_{A,b}(\mathbf{R}^n) \cap \operatorname{dom} g = \mathbf{R}^m \neq \emptyset$, Theorem (Composition with Affine Map) implies that the composition $\phi_{A,b} = g \circ L_{A,b}$ is convex for each fixed $(A, b) \in \mathcal{I}$.

By definition, f is the pointwise supremum of the family $\{\phi_{A,b}\}_{(A,b)\in\mathcal{I}}$, i.e.,

$$f(x) = \sup_{(A,b)\in\mathcal{I}} \phi_{A,b}(x). \tag{6}$$

We verify the hypotheses of Theorem (Pointwise Supremum):

- Each $\phi_{A,b}$ is convex, as shown above.
- The supremum is not identically $+\infty$. Since $\mathcal{A}$ and $\mathcal{B}$ are bounded, there exist constants $L_A, L_B > 0$ such that $\|A\| \leq L_A$ for all $A \in \mathcal{A}$ and $\|b\| \leq L_B$ for all $b \in \mathcal{B}$. Then, for any fixed $x \in \mathbf{R}^n$,

$$\|Ax - b\|^2 \leq \Big[\|A\| \cdot \|x\| + \|b\|\Big]^2 \leq \Big[L_A\|x\| + L_B\Big]^2 < +\infty, \tag{7}$$

  so, by the Triangle Inequality, the supremum over $\mathcal{I}$ is finite.

Therefore, Theorem (Pointwise Supremum) implies that f is convex. It should be noted that this theorem requires only that each member of the family $\{\phi_{A,b}\}_{(A,b)\in\mathcal{I}}$ is convex. Thus, no assumption on the index set $\mathcal{I} = \mathcal{A} \times \mathcal{B}$ is needed. In particular, the result holds regardless of whether $\mathcal{A}$ and $\mathcal{B}$ are convex. ☺

2nd Problem

34. ***Channel capacity.*** A key result in information theory establishes that the capacity of an additive white gaussian noise (AWGN) channel with bandwidth W, input power P, and noise variance σ^2, is given by

$$C = W \log\left(1 + \frac{P}{\sigma^2 W}\right).$$

Show that the capacity is a concave function of (P, W), i.e., show that $C : \mathbf{R}_{++} \times \mathbf{R}_{++} \to \mathbf{R}$, with $C(P, W)$ as in above, is concave. The constant $\sigma^2 > 0$ is given.

Solution to the 2nd Problem. In order to prove that C is concave, we apply Theorem (Second-order Criterion), i.e., C is concave if and only if $\nabla^2 C \preceq 0$ for all $(P, W) \in \mathbf{R}_{++} \times \mathbf{R}_{++}$.

To simplify the computation, define the auxiliary variable and its partial derivatives

$$\varrho := 1 + \frac{P}{\sigma^2 W}, \quad \frac{\partial \varrho}{\partial P} = \frac{1}{\sigma^2 W}, \quad \frac{\partial \varrho}{\partial W} = -\frac{P}{\sigma^2 W^2}. \tag{8}$$

so that $C(P, W) = W \log(\varrho)$ and $\varrho > 1$ for all $(P, W) \in \mathbf{R}_{++} \times \mathbf{R}_{++}$.

With this in mind, the first- and second-order derivatives of C are written as follows.

- The first-order partial derivatives of C are

$$\frac{\partial C}{\partial P} = \frac{W}{\varrho} \cdot \frac{1}{\sigma^2 W} = \frac{1}{\sigma^2 \varrho}, \tag{9}$$

$$\frac{\partial C}{\partial W} = \log(\varrho) + \frac{W}{\varrho} \cdot \left(-\frac{P}{\sigma^2 W^2}\right) = \log(\varrho) - \frac{P}{\sigma^2 W \varrho} = \log(\varrho) - \frac{\varrho - 1}{\varrho}, \tag{10}$$

  where the last equality uses the identity $P/(\sigma^2 W) = \varrho - 1$.

- The second-order partial derivatives follow from (9) and (10),

$$\frac{\partial^2 C}{\partial P\, \partial P} = -\frac{1}{\sigma^2 \varrho^2} \cdot \frac{\partial \varrho}{\partial P} = -\frac{1}{\sigma^4 W \varrho^2}, \tag{11}$$

$$\frac{\partial^2 C}{\partial P\, \partial W} = -\frac{1}{\sigma^2 \varrho^2} \cdot \frac{\partial \varrho}{\partial W} = -\frac{1}{\sigma^2 \varrho^2} \cdot \left(-\frac{P}{\sigma^2 W^2}\right) = \frac{P}{\sigma^4 W^2 \varrho^2}, \tag{12}$$

$$\frac{\partial^2 C}{\partial W\, \partial W} = \frac{\partial \varrho}{\partial W} \cdot \left(\frac{1}{\varrho} - \frac{1}{\varrho^2}\right) = -\frac{P}{\sigma^2 W^2} \cdot \frac{\varrho - 1}{\varrho^2} = -\frac{P^2}{\sigma^4 W^3 \varrho^2}. \tag{13}$$

The Hessian matrix is then given by

$$\nabla^2 C = \begin{bmatrix} -\dfrac{1}{\sigma^4 W \varrho^2} & \dfrac{P}{\sigma^4 W^2 \varrho^2} \\ \dfrac{P}{\sigma^4 W^2 \varrho^2} & -\dfrac{P^2}{\sigma^4 W^3 \varrho^2} \end{bmatrix}. \tag{14}$$

Factoring out the common scalar $-1/(\sigma^4 W \varrho^2)$ from (14), it is possible to obtain

$$\nabla^2 C = -\frac{1}{\sigma^4 W \varrho^2} \begin{bmatrix} 1 \\ -P/W \end{bmatrix} \begin{bmatrix} 1 & -P/W \end{bmatrix} = -\frac{1}{\sigma^4 W \varrho^2}\, v v^T, \quad v := [1,\ -P/W]^T \in \mathbf{R}^2. \tag{15}$$

Figure 1 illustrates $C(P, W)$ for $\sigma^2 = 1$, where the downward curvature of the surface suggests $\nabla^2 C \preceq 0$.

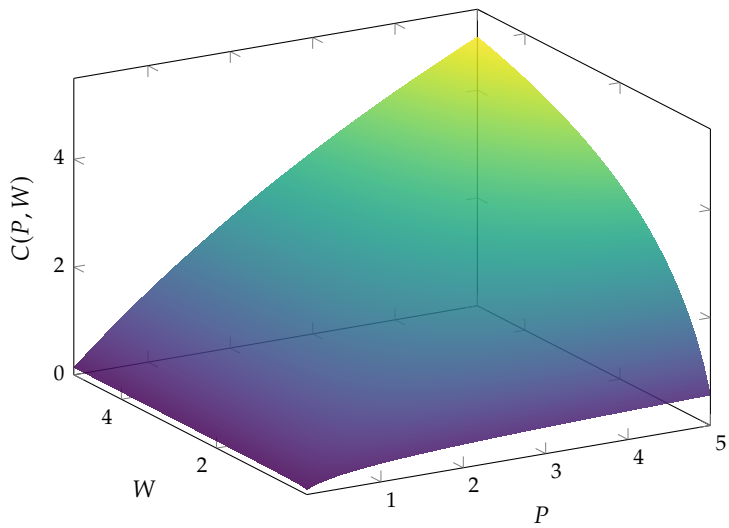

Figure 1: Capacity $C(P, W) = W \log(1 + P/W)$, for $\sigma^2 = 1$, illustrating the concavity of C over $\mathbf{R}_{++} \times \mathbf{R}_{++}$.

Now let us verify that in fact $\nabla^2 C \preceq 0$. From (15), this amounts to checking the sign of the scalar and the semi-definiteness of matrix vv^T:

- The matrix vv^T is positive semi-definite for any $v \in \mathbf{R}^2$, since $x^T(vv^T)x = (v^T x)^2 \geq 0$ for all $x \in \mathbf{R}^2$.
- The scalar $-1/(\sigma^4 W \varrho^2)$ is strictly negative, since $\sigma^2 > 0$, $W > 0$, and $\varrho > 0$.

This proves $\nabla^2 C \preceq 0$ for all $(P, W) \in \mathbf{R}_{++} \times \mathbf{R}_{++}$. By Theorem (Second-order Criterion), C is concave. ☺

3rd Problem

46. ***Concave function.*** Show that the function $f : \mathbf{S}^n_{++} \to \mathbf{R}$,

$$f(X) = \operatorname{tr}\left(X^{1/2}\right)$$

is concave. Recall that, if $X \in \mathbf{S}^n_{++}$ and $X = Q\Lambda Q^T$ denotes an eigenvalue decomposition of X, then $X^{1/2} = Q\Lambda^{1/2}Q^T$; here, $\Lambda^{1/2}$ is the diagonal matrix obtained by applying the square-root to each diagonal entry of the diagonal matrix Λ.

Hint: you can use the fact that, for $X \succ 0$, there holds

$$\operatorname{tr}\left(X^{1/2}\right) = \inf_{U \succ 0} \operatorname{tr}(XU) + \frac{1}{4}\operatorname{tr}(U^{-1}).$$

Solution to the 3rd Problem. In order to prove that f is concave, we use the given identity. For $X \succ 0$,

$$\operatorname{tr}(X^{1/2}) = \inf_{U \succ 0} \left[\operatorname{tr}(XU) + \tfrac{1}{4}\operatorname{tr}(U^{-1})\right]. \tag{16}$$

For each fixed $U \succ 0$, define the function

$$\phi_U : \mathbf{S}^n_{++} \to \mathbf{R}, \quad \phi_U(X) = \operatorname{tr}(XU) + \tfrac{1}{4}\operatorname{tr}(U^{-1}). \tag{17}$$

We claim that ϕ_U is both convex and concave for each fixed $U \succ 0$. To see this, note that the first term $\operatorname{tr}(XU) = \langle U, X\rangle_F$ is a linear function of X, where $\langle \cdot, \cdot \rangle_F$ denotes the Frobenius inner product, and the second term $w := \frac{1}{4}\operatorname{tr}(U^{-1}) \in \mathbf{R}$ is constant in X. Hence, $\phi_U(X) = \langle U, X\rangle_F + w$ is an affine function of X.

- By Theorem (Affine Functions), ϕ_U is convex for each fixed $U \succ 0$.
- Moreover, $-\phi_U$ is also affine, since the negative of an affine function is still affine. Therefore, by the same theorem, $-\phi_U$ is convex for each fixed $U \succ 0$.
- By the definition of concavity, this implies that ϕ_U and $-\phi_U$ are also concave for each fixed $U \succ 0$.

Thus, ϕ_U is both convex and concave for each fixed $U \succ 0$.

Furthermore, from (16), f is the pointwise infimum of the family $\{\phi_U\}_{U \succ 0}$, i.e.,

$$f(X) = \inf_{U \succ 0} \phi_U(X). \tag{18}$$

Therefore

$$-f(X) = -\inf_{U \succ 0} \phi_U(X) = \sup_{U \succ 0} (-\phi_U(X)). \tag{19}$$

We verify the hypotheses of Theorem (Pointwise Supremum):

- Each $-\phi_U$ is convex, as shown above.
- The supremum is not identically $+\infty$. For any $X \succ 0$, the eigenvalues of X are strictly positive. Taking the eigenvalue decomposition $X = Q\Lambda Q^T$ with Q orthogonal,

$$f(X) = \operatorname{tr}\left(X^{1/2}\right) = \operatorname{tr}\left(Q\Lambda^{1/2}Q^T\right) = \operatorname{tr}\Big(\underbrace{Q^TQ}_{I_n}\,\Lambda^{1/2}\Big) = \operatorname{tr}\left(\Lambda^{1/2}\right) = \sum_{i=1}^{n}\sqrt{\lambda_i} > 0, \tag{20}$$

  which gives $-f(X) < +\infty$.

Therefore, Theorem (Pointwise Supremum) implies that $-f$ is convex. By definition, f is concave. ☺